

- 1.6. Para el esquema de la base de datos Chinook hacer en AR y CRT una consulta que devuelva los nombres de los clientes que tengan la factura (invoice) con el ítem (invoice line) de mayor cantidad.

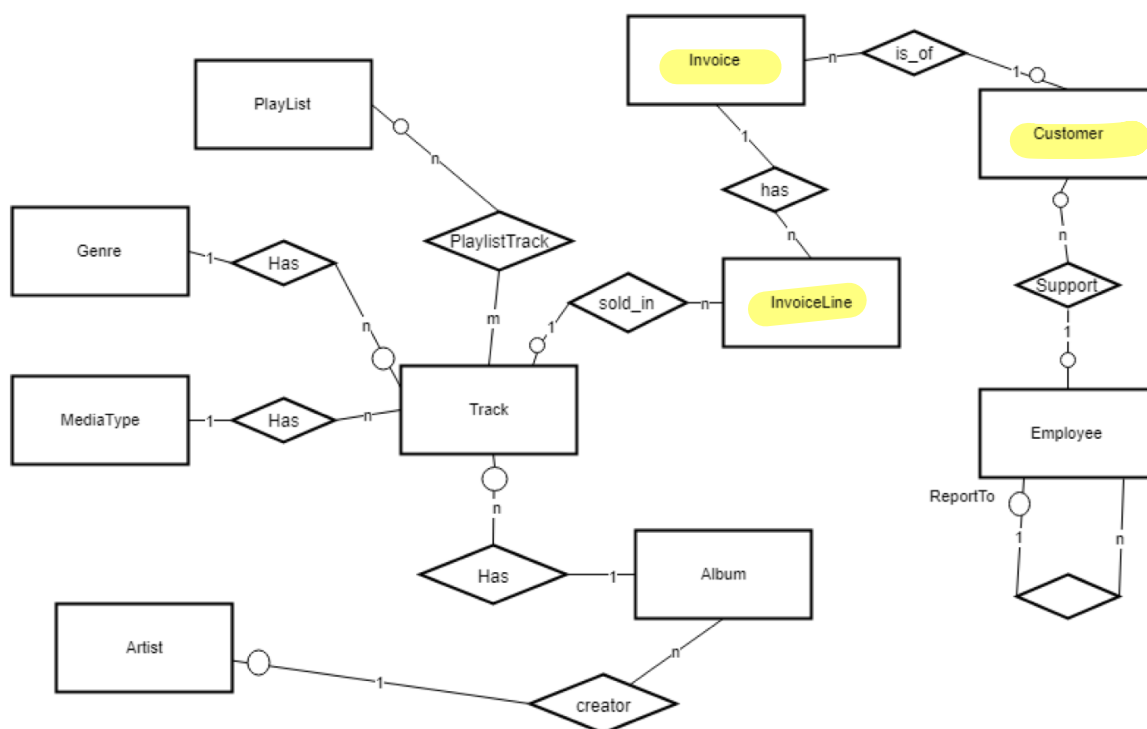


Figure 1: DER

artist (ArtistId, Name)
employee (EmployeeId, LastName, FirstName, Title, ReportsTo, BirthDate, HireDate, Address, City, State, Country, PostalCode, Phone, Fax, Email)
genre (GenreId, Name)
mediatype (MediaTypeId, Name)
playlist (PlaylistId, Name)
album (AlbumId, Title, ArtistId)
customer (CustomerId, FirstName, LastName, Company, Address, City, State, Country, PostalCode, Phone, Fax, Email, SupportRepId)
invoice (InvoiceId, CustomerId, InvoiceDate, BillingAddress, BillingCity, BillingState, BillingCountry, BillingPostalCode, Total)
track (TrackId, Name, AlbumId, MediaTypeId, GenreId, Composer, Milliseconds, Bytes, UnitPrice)
invoice line (InvoiceLineId, InvoiceId, TrackId, UnitPrice, Quantity)
playlist track (PlaylistId, TrackId)

AR

$\rho(IL_1, \text{Invoice Line})$

$\rho(IL_2, \text{Invoice Line})$

$\rho(\text{ menor-Cant.Items}, IL_1 \bowtie IL_2 \mid IL_1.qty < IL_2.qty)$

$\rho(\text{Items-Mayor-Cant}, \text{Invoice Line} - \text{menor-Cant.Items})$

Inc

$\rho(\text{Facturas-Mayor-Cont}, \text{Invoice} \times \text{IMC})$
 FMC
 Invoice.InvoiceId
 IMC.InvoiceId

$\rho(\text{Clientes-Mayor-Fact}, \text{Customer} \times \text{FMC})$
 CMF
 $\text{Customer.CustomerId}$
 FMC.CustomerId

$\rho(\text{Result}, \Pi_{\text{First-Name}, \text{LastName}}(\text{CMF}))$

CRT:

$\{t \mid \exists I, IL, C \mid$
 $IL \in \text{InvoiceLine} \wedge$
 $I \in \text{Invoice} \wedge$
 $C \in \text{Customer} \wedge$
 $\text{tieneMaxQty}(IL) \wedge$
 $I.\text{InvoiceId} = IL.\text{InvoiceId} \wedge$
 $C.\text{CustomerId} = I.\text{CustomerId} \wedge$
 $t.\text{Nombre} = C.\text{FirstName} \wedge$
 $t.\text{Apellido} = C.\text{LastName}$

$\text{tieneMaxQty}(IL) = \forall IL' \mid$
 $(IL' \in \text{InvoiceLine} \Rightarrow$
 $IL.\text{qty} \geq IL'.\text{qty})$